



Prepared with Synoptic

Final investigation report

Rejected take-off and runway overrun, Tarlton Springs

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Executive summary

What happened

On 18/06/2026, Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, began its take-off roll on runway 09 at Tarlton Springs Aerodrome at 17:30 local (07:30:02 UTC). This was five minutes after last light, with no moon. At 116 kt the first officer saw two kangaroos crossing onto the runway. The pilot in command rejected the take-off at 118 kt, 6 kt above V1, and the aircraft struck both animals about one second later. It left the runway end at approximately 21 kt and stopped upright on the extended centreline, 62 m into the 90 m runway end safety area and 28 m short of ungraded terrain. On board were 44 passengers, 2 flight crew and 1 cabin crew.

There were no serious injuries and no evacuation. One cabin crew member reported bruising while disembarking by the forward airstair. The aircraft was substantially damaged. Three No. 2 propeller blades were beyond limits, the right main gear doors were distorted, both right main tyres deflated through the fuse plugs and the lower right fuselage was punctured. Both kangaroos were killed. The aerodrome was closed for 4 hours 20 minutes.

What the investigation found

The stop began with 695 m of runway remaining. The strike removed No. 2 reverse, leaving deceleration about 15% below the certified value, and the stop needed 757 m. Neither factor alone would have caused the overrun.

The pilot in command judged that rotating into two large animals at night was the worse risk. The operator's training and manual gave no scenario or guidance for an obstruction during the take-off roll, and this likely contributed to a decision made in about a second. The pilot in command stated he knew the rule and departed from it deliberately. The gap explains the conditions of the decision rather than excusing it.

The kangaroos very likely entered through a northern boundary fence that the aerodrome operator had recorded as failed in 04/2023 and deferred through three budget rounds. The failed fence likely contributed. Because the aerodrome was below the standard's threshold for a documented wildlife plan, the fence was treated as an ordinary works item. This likely contributed to it staying unrepaired, although on a narrower balance of evidence than the other factors. The departure fell in the period holding nearly all kangaroo activity in the aerodrome's log, which likely contributed. Darkness likely contributed to the reject, because the strip either side of the seal could not be seen. The operator introduced the evening service without reopening a route risk assessment that restricted Tarlton Springs to daylight, which likely contributed to the roll being made in darkness.

No runway inspection was requested before departure, which likely contributed. The operator required one after last light only where a company wildlife strike had been reported within 24 months. That criterion very likely contributed, as Tarlton Springs was not listed despite eleven logged sightings of kangaroos on the sealed surface.

Maximum braking was established 2 seconds after the decision, against a fleet mean of 3.2 seconds. This very likely kept the aircraft within the prepared safety area.

A tailwind and a No. 2 engine power loss before the reject were ruled out. Crew fatigue was about as likely as not, and recorded performance showed no sign of impairment. Four northern edge lights were above nominal intensity, and whether this affected when the animals were seen could not be determined.

What can be done

On 26/06/2026 the operator listed Tarlton Springs by temporary revision TR-2026-04. The investigation recommends that the operator list aerodromes on wildlife exposure rather than on reported strikes, and that it adopt a route risk assessment trigger for any aerodrome first operated into after last light.

The aerodrome operator installed a temporary electric wire and programmed the fence replacement for 09/2026. [NOT ESTABLISHED: whether the replacement fence will exclude kangaroos.]

No action had been recorded on the standard's threshold for a wildlife plan, the change approval process, or training and guidance for an obstruction during the take-off roll.

Two further issues did not contribute. The evening roster closed five-sector duties within 45 minutes of the limit, which likely increased risk. Floor escape path lighting was checked only every 24 months, which concealed a defect on VH-SBQ and very likely increased risk. No action had been recorded on either issue.

Factual information

History of the flight

On 18/06/2026, Saltbush Air SB512, registered VH-SBQ, was scheduled to fly from Tarlton Springs Aerodrome to Port Carrick. It was the last scheduled sector of the day. On board were 44 passengers, 2 flight crew and 1 cabin crew. It was the fifth sector of the flight crew's duty. The crew had been assigned the evening rotation that morning, after a member of the rostered crew reported unfit.

Tarlton Springs has no control tower. The crew made their own broadcasts on the CTAF. They switched on the pilot-activated runway lighting at 07:21:50 and broadcast that they were taxiing at 07:23:15. The doors had closed at 07:16 (17:16 local), and the aircraft departed on schedule. The aerodrome was in darkness. The only runway inspection that day had been carried out at 21:35 on 17/06/2026 (07:35 local on 18/06/2026), before the first scheduled arrival, and it recorded the movement area as clear. The crew did not request a further inspection and the aerodrome did not offer one. No wildlife transmission was heard on the CTAF in the 90 minutes before the departure.

At 07:29:10 the pilot in command broadcast that SB512 was lining up on runway 09. The first officer was pilot flying. Power was set at 07:30:02. The roll began five minutes after last light, with no moon. The pilot in command called "Eighty knots" at 07:30:15 and "V one" at 07:30:29, at 112 kt.

At 07:30:31, at 116 kt, the first officer saw two kangaroos moving onto the runway from the northern side, beyond the 1,100 m point, and called them as being on the left. She later estimated their range at first sight as about 200 m, possibly less. At 07:30:32 the pilot in command called "Stopping. I have control." and rejected the take-off at 118 kt. This was six knots and 3 seconds past V1, with 695 m of sealed runway remaining. About one second later the aircraft struck both animals. The impact was taken by the right main landing gear and the No. 2 propeller.

Maximum braking was established at 07:30:34. The pilot in command selected reverse at 07:30:36, and only No. 1 reverse engaged. At 07:30:47 he said "We're not stopping on it." The aircraft left the end of runway 09 at approximately 21 kt at 07:30:51. It came to rest at 07:30:57, 62 m into the 90 m runway end safety area (RESA), upright and on the extended centreline, with the nose wheel in graded gravel.

The pilot in command set the parking brake and shut down both engines. There were no fire indications, and he did not command an evacuation. He told the cabin to remain seated. At his request on the CTAF, the aerodrome reporting officer activated the aerodrome emergency plan at 07:32:10 and sent a vehicle. At 07:32:24 the officer broadcast that the runway was obstructed and the aerodrome closed. Passengers were disembarked by the forward airstair, which was heard operating at 07:34:40. No slides were deployed and the main cabin lights stayed on. The occurrence notification records the disembarkation as complete 4 minutes 8 seconds after the aircraft stopped.

There were no serious injuries. One cabin crew member reported bruising during the disembarkation. Both kangaroos were killed. The aerodrome was closed for 4 hours 20 minutes. Both flight crew were drug and alcohol tested and both results were negative.

Personnel information

The pilot in command held 8,410 hours total, including 2,140 hours on type, and had been in command for 3 years 1 month. His last cyclic simulator check, on 11/03/2026, and his last line check, on 02/12/2025, were both satisfactory. He had no prior occurrence history. He had two rest days before the occurrence. His duty and rest in the preceding 7 days were within company limits, with no variations applied.

The first officer held 1,655 hours total, including 405 hours on type. Her last cyclic simulator check, on 28/04/2026, was satisfactory. She had flown into Tarlton Springs twice before, on 09/01/2026 and 17/02/2026, both times in daylight. She had one rest day before the occurrence.

The crew signed on at 06:30 local. They flew Port Carrick to Hensby, Hensby to Mardle Creek and Mardle Creek to Port Carrick as rostered, then SB511 and SB512. A ground delay in Port Carrick was recovered on the Tarlton Springs turnaround. The occurrence was 11 hours 00 minutes into a planned duty of 13 hours 00 minutes. The maximum for five sectors with that sign-on, under the operator's approved fatigue risk management system, was 13 hours 30 minutes.

At interview, both pilots described themselves as tired. The pilot in command said he was not struggling, and that he would have declined a further sector. The first officer described it as a normal five-sector day and said she had taken a proper break in Port Carrick.

Aircraft and equipment information

VH-SBQ was a 50-seat twin-turboprop. The take-off was planned at 20,800 kg with flap 10. [NOT ESTABLISHED: aircraft type, maintenance status and recorded serviceability before the occurrence.]

Meteorological information

The AWIS gave the wind as 090 degrees at 6 kt. V1 was calculated for a 5 kt headwind component on a dry runway. The headwind component recorded at the aircraft during the roll was 4 to 6 kt throughout. This was consistent with the 07:00 UTC routine observation. No tailwind component was recorded.

Recorded data

The FDR, CVR and CTAF recording were available. The FDR and CVR were removed at 12:40 UTC on 18/06/2026.

Both engines were at take-off torque (98.4% and 98.5%) from power set until the power levers were retarded at 07:30:32. No caution, warning or engine parameter exceedance was recorded before that point. Fuel flow, ITT and propeller RPM on both engines were within normal scatter. The power levers reached flight idle within 0.6 seconds. At impact, the FDR recorded a lateral acceleration spike of 0.31 g right and a vertical spike of 1.48 g. Maximum braking (2,940 psi) was established 2 seconds after the decision, with anti-skid active on all four wheels. The operator's fleet mean from decision to maximum braking is 3.2 seconds.

No. 2 propeller RPM decayed from 07:30:33, and the No. 2 reverse interlock did not make. Mean deceleration between 07:30:34 and 07:30:44 was 2.6 m/s². This is approximately 15% below the value used in the certified accelerate-stop data, which assumes symmetric reverse. Deceleration fell to 1.9 m/s² in the gravel.

Reconstruction from groundspeed placed the aircraft 1,152 m along runway 09 at the reject. The stopping distance achieved from 118 kt was 757 m.

After the aircraft stopped, the first officer said on the CVR that the animals "came out of the dark on the left of the strip, crossed the centreline." The pilot in command said "I never had them. I was heads-in on the engine page after V one."

Aerodrome and site information

Runway 09 has a 90 m RESA. Beyond the RESA the terrain is ungraded. The aircraft stopped 28 m short of the end of the prepared area.

The survey found two carcasses, at 1,188 m and 1,221 m from the runway 09 threshold. Tyre marks ran continuously from 1,160 m to the runway end, and the nose wheel rut ran to 62 m beyond it.

The aerodrome kept a paper wildlife sighting log covering 01/2025 to 06/2026. It held 46 entries. Of those, 41 recorded kangaroos on or adjacent to the movement area, and 38 of the 41 were timed between one hour before and two hours after last light. Eleven entries recorded animals on the sealed surface of runway 09/27. The log was not reported to any operator. The aerodrome reporting officer stated that no airline had ever asked for it. The northern fence had failed and had not been repaired through three budget rounds. The aerodrome had no documented wildlife plan. [NOT ESTABLISHED: fence location and condition, and the standard and threshold applying to a wildlife plan.]

Four northern edge lights were above nominal intensity on the night. [NOT ESTABLISHED: the measured intensities.] The pilot in command described the lighting as bright, with darkness either side of the lit corridor. The first officer described it as normal. She qualified this herself, noting that her only previous visits had been in daylight. She also said she could see the animals clearly once they were on the sealed surface.

Organisational and management information

The operator's Part A 8.14 and its Appendix 8A list the aerodromes at which a runway inspection is requested before the first departure after last light. An aerodrome is listed only where a wildlife strike has been reported there within 24 months. Tarlton Springs was not listed. The first officer had checked the flight plan, the charts, the aerodrome directory entry and the company aerodrome brief page before the flight, and none of them mentioned animals. The pilot in command had seen kangaroos on the strip in the evening four or five times. He had mentioned this to ground staff but had not reported it to the operator.

The evening rotation began on 14/07/2025 and operated on 289 days to 18/06/2026. On each of those days it was rostered as the fourth and fifth sectors of a duty signing on between 06:00 and 07:00. Those duties were planned at 12 hours 45 minutes to 13 hours 15 minutes, against the 13 hours 30 minute maximum. On six days before the occurrence, the rotation had been reassigned to a crew already on duty. The service was approved without reopening the earlier daylight-only risk assessment. Neither pilot recalled any briefing accompanying its introduction.

Both pilots stated that every rejected take-off scenario in their simulator training had been an engine event below V1. Neither had been given training or guidance on a runway obstruction during the take-off roll.

Tests and research

The No. 2 propeller had three blades damaged beyond repairable limits, including one blade tip separation. Animal tissue was recovered from two blade roots. A borescope inspection of both engines found no pre-existing damage and no evidence of a power anomaly before the impact. Both right main tyres had deflated through the fuse plugs. Brake energy absorbed was calculated at 84% of the certified maximum. The right main gear doors were distorted, but the gear structure was within limits. The lower right fuselage had abrasion and two punctures consistent with propeller debris. The anti-skid system tested serviceable.

The first officer reported unlit sections of the aisle floor strip, towards the rear, during the disembarkation. A functional check was carried out on 24/06/2026. [NOT ESTABLISHED: result of the check.] Floor escape path lighting was checked every 24 months and was not checked pre-flight.

Analysis

Standard's threshold left the aerodrome with no documented wildlife plan

Tarlton Springs is a certified aerodrome, and AS-139 section 10.4 governs how such an aerodrome manages wildlife. Section 10.4.1 obliges every certified aerodrome operator to "take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome." Section 10.4.2 requires a documented wildlife hazard management plan only where the aerodrome is used by aeroplanes with a maximum take-off weight "exceeding 27,000 kg", or serves "more than 350,000 passenger movements in a calendar year."

Tarlton Springs met neither limb. Its heaviest scheduled aeroplane was the 22,900 kg twin-turboprop used on the Saltbush Air service. It handled 61,400 passenger movements in 2025. Under 10.4.3, its measures therefore "need not be documented".

The aerodrome held no plan, no hazard assessment and no dispersal programme. Beyond two items, it had no record of the measures it took:

- the single inspection before the first arrival, under Aerodrome Manual 6.3
- a paper sighting log.

That position was confirmed against the certified aerodrome register and the aerodrome operator's own compliance file. The aerodrome was not in breach of the standard. The plan was missing because of where the standard set its threshold, not because the aerodrome failed to comply. It is virtually certain that the aerodrome had no documented wildlife plan.

The question is whether that absence contributed to the occurrence. The aerodrome did hold information about its wildlife hazard. The sighting log contained 46 entries between 01/2025 and 06/2026. Of these, 41 recorded kangaroos on or next to the movement area. Of those 41, 38 fell between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was kept in the aerodrome office and was not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked for it.

The council's works register shows the same pattern. Item WR-221, raised in 04/2023, recorded the northern fence between the 700 m and 1,300 m runway markers as failed, with animals observed passing through. It estimated replacement at \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in 02/2026, recorded the same section as deteriorating further.

Nothing in that history treated the fence as a wildlife control. It sat in the works register as an ordinary maintenance item competing for council money. Each deferral was made without a hazard assessment before the council. A documented plan brings a hazard assessment, a named owner and an audit against both. Under a plan, the fence would have been a recorded wildlife control, and its failure would have had to be answered for. It would not have been a works item that could simply be deferred.

The evidence against this argument is real. Section 10.4.1 still required reasonable measures, so the absence of a plan did not remove the obligation to act. A documented plan is not the only way to repair a fence or to pass a sighting log to an airline. The deferrals were budget decisions, and no evidence showed that a fence named in a plan would have been funded any better. A plan does not of itself pay for a fence.

The investigation weighed these arguments and found the case for contribution the stronger. The deciding point was that the council deferred the repair three times without the hazard assessment a plan would have put before it. On that basis, the absence of a plan likely contributed. The investigation holds this finding with less confidence than its other findings on contribution, and it

is not presented as settled. The mechanism is inferred. No evidence shows what the council would have decided with a hazard assessment in front of it.

The absence of a plan reached the occurrence only through the failed northern fence. The investigation found that the failed fence contributed to the kangaroos crossing onto the runway, and that factor is argued under organisational influences. The absence of a plan was not found to bear on the operator's criterion for a runway inspection after last light. That criterion is the operator's own rule at Appendix 8A, and no aerodrome plan could have changed it.

The standard's threshold left the aerodrome without a documented wildlife plan, and that absence likely contributed to the failed northern fence remaining unrepaired, although this rests on an inferred mechanism and a narrower balance of evidence than the other contributing factors.

Organisational Influences

Failed northern fence left unrepaired through three budget rounds

The aerodrome operator's works register recorded the northern boundary fence as failed in 04/2023. Item WR-221 described four displaced panels and strained wire between the 700 m and 1,300 m runway markers. It noted animals passing through and recommended replacement at an estimated \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263 recorded the same section deteriorating further in 02/2026. The site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts and an animal pad through it at the 1,050 m mark (Figure 1). It was virtually certain that the fence had failed and stayed failed, because this rests on the operator's own entries and was confirmed on the ground.



Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026

The pad lay on the same side of the runway as both carcasses. It was 138 m before the first, recovered at 1,188 m, and the second was at 1,221 m. The failed section was therefore the route by which these two animals reached the movement area. Had WR-221 been funded in 2023, that section would have been intact by 06/2026 and the route closed.

There was evidence against this. WR-221 stated no specification for the replacement, and kangaroos can clear a stock fence. The pad shows that these animals came through the gap. It does not show that an intact fence would have stopped them. The wildlife sighting log also does not show where any of its 41 kangaroos entered the aerodrome. This uncertainty limits the conclusion but does not remove it. The register itself linked the failure to animals passing through.

The northern fence, left unrepaired through three budget rounds, likely contributed to the two kangaroos reaching the sealed runway. Whether a replacement fence would have excluded them could not be determined.

Evening service approved without reopening the daylight-only risk assessment

RRA-041 issue 3 (11/2022) was the current route risk assessment for Tarlton Springs. It carried the restriction "daylight operations only pending review of night lighting and diversion fuel policy". CMR-2025-118 introduced the evening rotation from 14/07/2025. This was the first scheduled company operation into Tarlton Springs after last light. The change was assessed as "schedule change only", with "No new hazard identified", and CMR-2025-118 answered "Route risk assessment review required: No." The Head of Flying Operations approved it on 22/05/2025. CMR-2025-118 did not refer to the daylight restriction. RRA-041 was not reopened, and no issue 4 exists.

The Safety Department had no involvement beyond receiving the approved document. That the service was approved without reopening the assessment was virtually certain. The approver has since left the company and could not be interviewed. Why the change was assessed in this way was therefore not established, and the investigation drew no conclusion about intent.

Had the restriction been honoured, every company departure from Tarlton Springs would have been in daylight. SB512 was scheduled off blocks at 17:20 local and departed on schedule. It began its roll at 17:30, five minutes after the end of civil twilight at 17:25. The darkness of the roll came from the schedule, not from a delay.

The evidence against rests on what the restriction named. It referred to night lighting and diversion fuel, not wildlife. A review might have dealt with those two items and approved the service with conditions. The departure after last light would then have been unchanged. The investigation considered this the less likely outcome. Lifting a daylight-only restriction would have required assessing a night operation at an aerodrome with a single runway, no lighting redundancy and no alternate within 200 nm at night. Without the approver, what a review would have concluded remains an inference. The change process also could not have altered the Appendix 8A criterion, which is considered with the operator's risk controls. How the darkness affected the reject is considered with the local conditions.

Approving the evening service without reopening RRA-041 likely contributed to SB512 beginning its take-off roll in darkness.

Evening rotation rostered to close five-sector duties within 45 minutes of the limit

The evening rotation operated on 289 days between 14/07/2025 and 18/06/2026. On every one of them it was rostered as the fourth and fifth sectors of a Port Carrick-based duty signing on between 06:00 and 07:00. Those duties were planned at between 12 hours 45 minutes and 13 hours 15 minutes, against a maximum of 13 hours 30 minutes. On six of those days a rostered crew member reported unfit. Each time the rotation went to a crew already on duty, which extended that crew's day from three sectors to five. 18/06/2026 was the sixth such day. The crew signed on at 06:30 and began the take-off 11 hours into a planned 13 hour duty, within the company's limits. This roster pattern was virtually certain to have existed.

The roster placed the last sector of the longest duty in the pattern into an aerodrome with no alternate within 200 nm at night. On 18/06/2026 it gave that sector to a crew whose day had just been extended, and both crew members described themselves as tired. The recorded data did not show that this affected the occurrence. The power levers reached flight idle 0.6 seconds after the decision. Maximum braking followed at 2 seconds, against a fleet mean of 3.2 seconds. There was no omission during the roll. The one decision a long duty could plausibly have affected was not requesting a runway inspection, and no document prompted either crew member to consider one. Whether the crew were fatigued could not be resolved, and that matter is discussed separately. It was unlikely that the roster influenced the occurrence.

The roster pattern remains important for future operations. It exposes every operation of the rotation, and the exposure is greatest whenever a crew member reports unfit and the sectors pass to a crew already on duty.

Rostering the evening rotation to close five-sector duties within 45 minutes of the limit likely increased risk to the operation, although it was unlikely to have contributed to this occurrence.

Risk Controls

Inspection after last light required only where a wildlife strike was reported in 24 months

The operator had a defence aimed at this hazard. Part A 8.14.2 requires the pilot in command, at an aerodrome listed in Appendix 8A, to "request a runway inspection from the aerodrome operator prior to the first departure after last light". The pilot in command must also not start the roll until the result is passed. It was virtually certain that this requirement did not apply at Tarlton Springs. The manual text and the appendix listing are documentary, and the aerodrome had never been listed.

The control was sound. Its trigger was not. Appendix 8A admits an aerodrome only where "a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." Nothing in Part A asked a crew to report a sighting. Nothing admitted an aerodrome to the list on any other evidence. The aerodrome's own sighting log for 01/2025 to 06/2026 held 41 kangaroo entries. Of these, 38 fell between one hour before and two hours after last light, and eleven placed animals on the sealed surface of runway 09/27. None of those entries was a strike, and none reached the operator. The control could therefore be armed only by the outcome it existed to prevent.

Had Tarlton Springs been listed, the crew would have requested an inspection and would have needed a result before the roll. Aerodrome staff disperse animals when they find them. The assessment stops short of certainty for one reason. An inspection completed some minutes before departure does not guarantee a clear runway at the moment of the roll. The crew's not requesting an inspection was itself a contributing factor, and it is analysed under Individual Actions.

The listing criterion in Appendix 8A very likely contributed to the kangaroos being on the runway during the roll, because it left the operator's inspection requirement unarmed at an aerodrome with a documented history of kangaroos on the sealed surface around last light.

No training or guidance for a runway obstruction during the take-off roll

It was extremely likely that neither training nor procedure addressed an obstruction during the take-off roll. Module 4 of TRG-SYL-07 scripts four scenarios, and all four are aircraft system events. None presents an animal or any external hazard on the runway, at any speed. The training records covered five cycles, and no crew in them had been given an external obstruction scenario. Records before those five cycles were not available. Part B 4.6 has the same gap. Paragraph 4.6.4 states that "At or above V1 the take-off shall be continued." The section has no guidance on an obstruction seen on the runway, and no amplification of 4.6.4 for a case in which continuing is itself hazardous. Both flight crew independently stated that every stop or go scenario they had been given was an engine event.

Two readings of the gap's effect were weighed. On the first, the captain had about a second to decide and no briefed or practised frame to decide against. The choice he made at 118 kt was the first time he had considered the question. An unrehearsed decision made in about a second is not the same decision as one made against a practised frame.

On the second reading, the gap made no difference. The captain stated that he knew the rule and departed from it deliberately because he judged continuing to be worse. Training would not have changed what he knew. On that reading, the gap is a finding about the fleet rather than a cause of this reject.

The investigation accepted the first reading as likely, while recognising the weight of the second. The gap explains the conditions under which the decision was made. It does not excuse the decision.

The absence of training and guidance for a runway obstruction during the take-off roll likely contributed to the take-off being rejected 6 kt above V1, because it left the captain to decide in about a second with no rehearsed frame to decide against.

Floor escape path lighting checked only every 24 months, not pre-flight

On this type, the pre-flight emergency lighting test illuminates the ceiling emergency lights and the exit signs. It neither illuminates nor monitors the floor proximity escape path segments. Those segments are checked only by a dedicated functional test at the maintenance programme's 24-month interval. It was virtually certain that this monitoring gap existed. The test coverage and the interval are documentary.

On VH-SBQ the gap concealed a defect. The first officer noticed unlit sections of the aisle floor strip during the disembarkation. The functional check on 24/06/2026 then found two of the six segments aft of row 16 on the left side dark on battery power, traced to a failed common ballast. The last functional test had been 11 months earlier. The defect could therefore have gone undetected for up to 13 more months, and no check available to the crew or a line engineer could have found it.

It was unlikely that the gap or the defect influenced the overrun or its outcome. No evacuation was commanded, and the main cabin lights were on throughout. The disembarkation by the forward airstair took 4 minutes 8 seconds without incident.

The risk lies in what the gap hides. A fire indication, smoke or a night evacuation over the wing could each make two dark aft segments the difference between an orderly exit and a blocked one. Between functional tests, the state of this lighting on any aircraft of the type is unknown. [NOT ESTABLISHED: whether the failed ballast part is fitted across the fleet, and how many aircraft are currently inside the test interval]

The absence of any pre-flight check of the floor proximity escape path lighting very likely increased risk, because a defect in it can remain undetected on any aircraft of the type for up to 24 months.

Local Conditions

Departure in the logged kangaroo peak, 1 h before to 2 h after last light

The aerodrome's wildlife sighting log held 41 kangaroo entries for 01/01/2025 to 18/06/2026. Of these, 38 fell between one hour before and two hours after last light. Eleven placed animals on the sealed surface of runway 09/27. On 18/06/2026 that window ran from 16:25 to 19:25 local, and the roll began at 17:30. The scheduled departure therefore fell inside the window regardless of how the day's timetable ran. The captain's account matched the log. He had seen kangaroos on the strip at Tarlton Springs four or five times, always in the evening.

The log has two limits. It is a paper register, and its completeness cannot be verified. It also records sightings rather than presence. Aerodrome staff go off duty at 18:30, part way through the window, so the pattern may partly reflect when someone was there to see. Neither limit overturns the pattern. The concentration of entries is pronounced, and the captain's experience is independent of the log. Two animals crossing the runway at 17:30 matched the pattern the log describes and was not a chance encounter.

It was extremely likely that SB512 departed within the aerodrome's logged period of peak kangaroo activity, and likely that this contributed to two kangaroos crossing onto the sealed runway during the roll.

Roll in darkness, 5 minutes after last light, with no moon

The roll began 32 minutes after sunset and 5 minutes after the end of civil twilight. Moonrise was not until 22:41. There was no town lighting in sight, and the only light came from the runway edge lighting. The investigator's site visit at the same time of day confirmed that the unlit graded strip either side of the seal cannot be seen from the centreline.

In these conditions a crew's first view of a crossing animal is the animal on the lit surface. The first officer's words on the CVR after the stop were that the animals "came out of the dark on the left of the strip". She said there was "nothing to see and then there were two of them". Her account describes animals arriving out of the dark, not animals she was slow to see. In daylight, animals crossing the graded strip from the northern fence would have been in view before they reached the seal.

Two points weigh against this. How long the animals were on the strip before reaching the seal is not known. Animals that entered the strip only seconds before reaching the seal would have been seen late in daylight too. The captain was heads-in on the engine page after V1, where the procedure places him. A daylight sighting would therefore still have rested on the pilot flying, whose attention belongs on the centreline. These points limit the strength of the argument but do not defeat it, because the site visit shows the strip was invisible in this light.

Departing after last light on schedule was the outcome of approving the evening service without reopening the daylight-only risk assessment. That matter is argued under Organisational Influences.

It was virtually certain that the roll took place in darkness with no moon, and likely that this delayed the sighting of the animals and so contributed to the take-off being rejected 6 kt above V1.

Four northern edge lights above nominal intensity on the night

A photometric check seven days after the occurrence measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal intensity for stage 3. They were in the section the animals crossed. The lighting runs at one fixed setting that crews cannot vary, and no work on it was recorded in between. It was therefore likely that the lights were at that intensity on the night, although nobody measured them at the time.

Whether that intensity delayed detection could not be settled. In support, the captain recalled the lighting as brighter than he would want, with everything outside the lit corridor black. Brighter edge lights would deepen the contrast with the dark strip from which the animals came. Against this, the first officer described the lighting as normal and saw the animals clearly once they were on the seal. She added that, with only two daylight visits to compare against, she was not well placed to judge. The captain's non-detection is explained by his heads-in duties after V1. The strip cannot be seen from the centreline in darkness whatever the edge lights are doing, and darkness alone accounts for nothing being seen off the seal.

Resolving the question would need evidence linking edge-light intensity of this order to the detection of an animal on or beside a runway, and none was available. The matter remains unresolved, and no conclusion is drawn from it.

Tailwind component during the take-off roll and stop

A tailwind was ruled out because the aerodrome's automatic observations and the recorded aircraft data both showed a headwind component throughout the roll and the stop.

Flight crew fatigued at the time of the take-off

The take-off was 11 hours into a duty that began at 06:30 local. It was the fifth sector, on a day extended that morning from three sectors to five. Both crew described themselves as tired. The captain said he would have declined another sector.

Against this, both described the tiredness as ordinary and not impairing. The duty was within the company's limits and followed rest days for both crew. The first officer had taken a break in Port Carrick during the day. The recorded performance showed none of the usual signs of impairment. The power levers reached flight idle 0.6 seconds after the decision, maximum braking was established at 2 seconds, the callouts were on time, and there was no omission on the roll.

Tiredness of this kind is not the same as fatigue that degrades performance, and no other evidence reached the question. It would have needed a detailed fatigue assessment, including sleep history and the duty pattern before 18/06/2026. Even if fatigue had been present, each mechanism through which it could have acted here was contradicted by recorded data. Whether the crew were fatigued remains unresolved, and nothing is carried from it into the conclusions. The risk posed by the duty itself is addressed under Organisational Influences.

Individual Actions

No runway inspection requested before departing after last light

It is virtually certain that no runway inspection was requested before SB512 departed. Two independent recordings establish this. Neither the CVR nor the CTAF recording contains a request for an inspection or an offer of one. Neither mentions wildlife at any point before the roll. The aerodrome's daily inspection log shows one inspection that day, at 07:35 local before the first arrival, and none afterwards. Both crew members said they did not consider asking for one.

The inspection was available. Aerodrome Manual 6.3 provides for an inspection outside the scheduled time at the request of a pilot in command. The reporting officer was on duty at the terminal throughout the turnaround and confirmed that an inspection would have been carried out if asked. SB512 was the only departure after last light that day, so no other traffic competed for it.

Nothing prompted the crew to ask. Part A 8.14.2 requires the request only at aerodromes listed in Appendix 8A, and Tarlton Springs had never been listed. At all other aerodromes, 8.14.3 states that "no specific action is required." The captain had expected Tarlton Springs to be on the list and was surprised to find it was not. No document asked either pilot to consider an inspection. The omission therefore had no owner. Why the list could not include Tarlton Springs is argued under Risk Controls.

An inspection would have covered the section of runway between 1,050 m and 1,250 m. The fence pad and both carcasses were found in that section. Aerodrome staff disperse animals when they find them. Even so, an inspection made some minutes before the roll could not guarantee that the runway would stay clear. For that reason the effect is not put higher.

It is likely that the absence of a runway inspection contributed to two kangaroos crossing onto the sealed runway during the take-off roll.

Take-off rejected 6 kt above V1

It is virtually certain that the take-off was rejected above V1. The FDR shows V1 at 112 kt at 07:30:29. Both power levers were retarded to flight idle from 07:30:32 at 118 kt, three seconds and six knots later. The CVR holds the decision in a single exchange. The first officer called the sighting at 07:30:31, and one second later the captain said "Stopping. I have control."

Part B 4.6.4 states that "At or above V1 the take-off shall be continued." The captain stated plainly that the reject was not by the book. In interview and in his air safety report, he described his reasoning. He had judged that rotating into two large animals would leave the aircraft airborne at night with damage he could not assess, over country offering no usable option. He chose the risk whose outcome he could see. Section 4.6 gave him no guidance to weigh that judgement against. It contains nothing on an obstruction on the runway at any speed, and that gap is argued under Risk Controls. The darkness and the presence of the animals are argued under Local Conditions. So is the brightness of four northern edge lights, and whether it delayed the sighting could not be determined.

The reject did not reduce the aircraft's deceleration. It reduced the runway left for the stop. The scheduled accelerate-stop distance of 1,780 m assumes that the reject begins at or before V1, and it left a margin of 67 m to the distance available. The reject began 1,152 m along runway 09, with 695 m of sealed surface remaining. The stop actually achieved from 118 kt took 757 m. Beginning six knots and three seconds late consumed the 67 m margin and more, before the loss of No. 2 reverse is counted. That loss belongs to the separate sequence from the strike to the reduced deceleration.

It is extremely likely that rejecting the take-off at 118 kt contributed to the overrun, because the stop began with less sealed runway than it needed.

Maximum braking established within 2 seconds of the reject decision

Recorded data establishes the crew's response to the reject as virtually certain. The power levers reached flight idle 0.6 seconds after the decision. Maximum braking was established at 07:30:34, 2 seconds after it. The operator's fleet mean from decision to maximum braking is 3.2 seconds. Anti-skid was active on all four wheels throughout. Brake energy reached 84% of the certified maximum without a brake fire. No. 1 reverse was selected at 07:30:36. No. 2 reverse was unavailable because its propeller had been damaged in the strike. The first officer made the speed callouts and the runway-end call.

The aircraft left the seal at 21 kt. It stopped upright on the extended centreline, 62 m into the 90 m RESA and 28 m short of its far end. Passengers were disembarked by the forward airstair without an evacuation. A stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m. That figure is to be read against the 28 m of RESA that remained.

The reject was outside procedure. The stop that followed it was executed faster than the fleet average.

It is very likely that establishing maximum braking within 2 seconds reduced how far the aircraft travelled beyond the runway end.

Occurrence Events

Two kangaroos crossed the strip onto the sealed runway

It is virtually certain that two kangaroos crossed the strip and reached the sealed surface of runway 09. Three separate pieces of evidence each establish this. Two adult carcasses were recovered from the seal at 1,188 m and 1,221 m from the threshold, on the northern half. The CVR records the first officer's sighting call at 07:30:31. The FDR records an impact at 07:30:33.

The first officer's account fits the physical evidence. After the aircraft stopped, the first officer described the animals coming out of the dark on the left of the strip and crossing the centreline. On runway 09 the left side is the northern side.

The carcass positions also bear on how the animals reached the strip. The animal pad through the failed northern fence is at the 1,050 m mark, 138 m before the first carcass and on the same side. Which route the animals took is a separate question. It is argued with the fence under Organisational Influences, and nothing in this subsection depends on it.

The animals' presence led directly to both the reject and the strike. The first officer's call is the only trigger for the reject on the CVR. The captain's "Stopping. I have control." followed one second later. No engine call, caution or warning came before it. The impact followed one second after the decision.

It is extremely likely that the animals' presence on the sealed runway led to both the rejected take-off and the strike.

No. 2 propeller and right main gear struck two kangaroos

It is virtually certain that the No. 2 propeller and the right main gear struck the animals. The impact was at 07:30:33 at 114 kt, one second after the reject began. The FDR recorded a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. Three No. 2 propeller blades were damaged beyond repairable limits, and one blade tip had separated. Animal tissue was recovered from two blade roots. The right main gear doors were distorted, and propeller debris abraded and punctured the lower right fuselage skin. The recorded data, the damage and the recovered tissue all agree.

The strike mattered most because of what it did to reverse thrust. No. 2 propeller RPM decayed from the moment of impact, and the No. 2 reverse interlock did not make. Reverse was selected at 07:30:36 on No. 1 only. The scheduled accelerate-stop figures assume symmetric reverse. The strike therefore removed half the available reverse thrust in the first seconds of a stop whose scheduled margin was 67 m.

It is virtually certain that the strike removed No. 2 reverse for the rest of the stop, which reduced the deceleration the aircraft could achieve.

Deceleration below the scheduled accelerate-stop performance

It is virtually certain that the aircraft decelerated more slowly than the certified accelerate-stop data assumes. Maximum braking was established at 07:30:34, with anti-skid active on all four wheels. The mean deceleration between 07:30:34 and 07:30:44 was 2.6 m/s². That is approximately 15% below the value used in the certified data for this weight and runway condition.

The braking system was not the source of the shortfall. The anti-skid system tested serviceable after the occurrence, and the brakes absorbed 84% of the certified maximum energy without fire or hydraulic leak. The shortfall coincided with the loss of No. 2 reverse.

The distance reconstruction shows what the shortfall cost. At the reject the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining. At the certified deceleration from that point, the stop would have taken roughly 660 m and ended on the runway. At the deceleration achieved, it took 757 m. The aircraft left the seal at about 21 kt and stopped 62 m into the RESA. The runway marks support the reconstruction. Tyre marks ran continuously from 1,160 m to the runway end, and the nose wheel rut ended 62 m beyond it.

The overrun had two causes acting on one stop. The reject began after V1, which shortened the runway left for the stop. That route is argued under Individual Actions. The strike lowered the deceleration achieved over the runway that remained. Neither contribution alone carried the aircraft past the runway end, but together they did.

It is virtually certain that the deceleration shortfall caused by the loss of No. 2 reverse contributed to the aircraft overrunning onto the runway end safety area.

No. 2 engine power loss before the reject decision

A No. 2 engine power loss before the reject decision was examined and found very unlikely: both engines held take-off torque within 0.1% of each other, with no caution, warning or exceedance, until the power levers were retarded at 07:30:32, and the No. 2 loss began only at the impact one second later.

Conclusions

These findings are made with respect to the occurrence. They should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. Tarlton Springs Aerodrome fell below both limbs of the AS-139 10.4.2 threshold for a documented wildlife hazard management plan, so it was not in breach of the standard. It held no plan, no hazard assessment and no dispersal programme. Without a plan, the failed northern fence was likely deferred as an ordinary works item, and the council considered it with no hazard assessment before it. A plan would not of itself have funded the repair. (Safety issue)
2. The northern perimeter fence between the 700 m and 1,300 m runway markers was recorded as failed in 04/2023, and animals were observed passing through it. The repair was then deferred unfunded at the 2023, 2024 and 2025 budget rounds. The fence's failure likely contributed to the two kangaroos reaching the runway through the animal pad at the 1,050 m mark, 138 m before the first carcass. It was not established, however, that the fence as replaced would have excluded kangaroos. (Safety issue)
3. The operator approved the evening service under CMR-2025-118 as a schedule change only. It did not reopen RRA-041 issue 3, which restricted Tarlton Springs to daylight operations. This likely resulted in SB512 rolling in darkness on its schedule. The approver could not be interviewed, and a review might have approved the service with lighting and fuel conditions. (Safety issue)
4. From 14/07/2025 to 18/06/2026, every operation of the evening rotation closed a five-sector duty that had been planned to within 45 minutes of the 13 hour 30 minute maximum. On six of those days, the rotation extended an on-duty crew's day from three sectors to five. This likely increased risk to the rotation's operations. It was not found to have contributed to the occurrence. (Safety issue)
5. Operations Manual Part A 8.14 required a runway inspection before the first departure after last light, but only at aerodromes listed in Appendix 8A. An aerodrome was listed only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. Tarlton Springs had therefore never been listed, despite eleven logged sightings of kangaroos on the sealed surface. This very likely resulted in no inspection being required or requested before SB512 departed. (Safety issue)
6. It is very likely that neither the operator's recurrent training nor Operations Manual Part B 4.6 contained any scenario or guidance for an obstruction on the runway during the take-off roll. That absence likely contributed to the reject decision being made without a briefed or practised frame. The pilot in command nonetheless knew the requirement to continue at or above V1, and he departed from it deliberately. (Safety issue)
7. On the type, the pre-flight emergency lighting test neither illuminated nor monitored the floor proximity escape path segments. Their only check was a functional test at a 24-month interval. On VH-SBQ, that functional check found two of the six segments aft of row 16 on the left side dark, on a failed common ballast. This very likely increased risk in any future evacuation. It was not found to have contributed to the occurrence or its outcome. (Safety issue)
8. The aerodrome's sighting log placed 38 of its 41 kangaroo entries in the period from one hour before to two hours after last light. It is very likely that SB512 departed within that period, and

departing then likely contributed to the two kangaroos being on the runway. The log recorded sightings rather than presence, and its completeness could not be verified.

9. The roll began 5 minutes after the end of civil twilight, with no moon. At that time, the graded strip either side of the sealed surface could not be seen from the centreline. The darkness likely delayed the point at which the crew could see the animals, and so contributed to the reject being made above V1. How long the animals had been on the strip before reaching the seal, however, was not known.

10. Four northern edge lights between the 900 m and 1,200 m markers were likely at 118% to 131% of nominal intensity on the night. Whether that intensity affected when the animals were seen could not be determined either way. The pilot in command described the lighting as brighter than he would want. The first officer, who saw the animals, described it as normal, and he saw them clearly once they were on the seal. Darkness alone accounted for nothing being seen off the seal, and no evidence linked an edge light at that intensity to the detection of an animal. The edge lights were not found to have contributed to the occurrence.

11. It is very unlikely that there was a tailwind component at any point in the roll or the stop. Both the aerodrome's automatic observations and the recorded aircraft data gave a headwind component throughout.

12. Whether the flight crew were fatigued at the time of the take-off could not be determined either way. Both pilots described themselves as tired 11 hours into the duty, but both also described the tiredness as ordinary and not impairing. The duty was within company limits, and the recorded responses on the roll were at the fast end of the fleet's distribution. Settling the question would have needed a detailed fatigue assessment, including sleep history and the duty pattern before 18/06/2026.

13. No runway inspection was requested before SB512 departed after last light, although the aerodrome reporting officer was on duty and would have carried one out if asked. Neither pilot considered an inspection, because no document prompted them to. The absence of an inspection likely contributed to the two kangaroos being on the runway at the roll. Even so, an inspection some minutes before departure could not have guaranteed a clear runway.

14. After the first officer's sighting call, the pilot in command rejected the take-off at 118 kt, which was 6 kt and 3 seconds above V1. He judged that continuing into two large animals at night carried the greater risk. Beginning the stop at that point very likely contributed to the overrun, because it left 695 m of sealed runway against a scheduled accelerate-stop margin of 67 m.

15. Maximum braking was established 2 seconds after the reject decision, against a fleet mean of 3.2 seconds. This very likely reduced the severity of the overrun. A stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m.

16. Two kangaroos crossed from the northern side onto the sealed surface of runway 09. They were recovered at 1,188 m and 1,221 m from the threshold. Their presence very likely led to both the reject and the strike.

17. One second after the reject began, at 114 kt, the No. 2 propeller and right main gear struck both kangaroos. The strike destroyed the No. 2 propeller and so removed No. 2 reverse thrust from the stop.

18. With reverse available on No. 1 only, the mean deceleration was 2.6 m/s², approximately 15% below the certified value. As a result, the aircraft needed 757 m to stop from the reject point, with only 695 m of runway remaining.

19. It is very unlikely that the No. 2 engine lost power before the reject decision. Until the power levers were retarded, both engines were at matched take-off torque, with no caution, warning or exceedance. The No. 2 power loss began at the impact one second later.

Probable cause

The overrun resulted from two causes acting together. The first was a take-off rejected 6 kt above V1 with 695 m of runway remaining, after two kangaroos had crossed onto the sealed runway. The second was the strike on those animals, which removed No. 2 reverse thrust and reduced the deceleration to approximately 15% below the certified value, so that the stop needed 757 m. The contribution of the late reject and of the animals' presence on the runway was very likely. The contribution of the strike and the reduced deceleration was virtually certain.

Contributing factors

- It is virtually certain that the AS-139 10.4.2 threshold left Tarlton Springs with no documented wildlife plan. It is likely that the absence of a plan contributed by leaving the northern fence to be deferred without a hazard assessment, though a plan would not of itself have funded the repair. (Safety issue)
- The failed northern fence was virtually certainly left unrepaired through three budget rounds. Its failure likely contributed by leaving open the route the kangaroos used. It was not established, however, that a replacement fence would have excluded kangaroos. (Safety issue)
- It is very likely that SB512 departed in the logged kangaroo peak, from 1 h before to 2 h after last light. It is likely that this contributed to the animals being on the runway.
- Under the operator's criterion, an inspection after last light was required only where a wildlife strike had been reported in the preceding 24 months. That criterion virtually certainly existed. It very likely contributed by leaving Tarlton Springs without any inspection requirement. (Safety issue)
- No runway inspection was requested before departing after last light. The absence of an inspection likely contributed to the kangaroos being on the runway when the roll began.
- Two kangaroos crossed the strip onto the sealed runway. Their presence very likely contributed to both the reject and the strike.
- It is very likely that no training or guidance existed for a runway obstruction during the take-off roll. It is likely that this gap contributed to the reject decision, although the pilot in command knew the requirement to continue at or above V1. (Safety issue)
- The evening service was virtually certainly approved without reopening the daylight-only risk assessment. This likely contributed by putting the roll in darkness, although what a review would have concluded could not be established. (Safety issue)
- The roll began in darkness, 5 minutes after last light, with no moon. The darkness likely contributed by delaying when the crew could see the animals.
- The take-off was rejected 6 kt above V1. The late start of the stop very likely contributed by leaving only 695 m of runway for it.
- With virtual certainty, the No. 2 propeller and right main gear struck two kangaroos. The strike virtually certainly contributed by removing No. 2 reverse thrust.
- Deceleration was virtually certainly below the scheduled accelerate-stop performance. That shortfall virtually certainly contributed to the aircraft leaving the runway end.

Other factors that increased risk

- It is virtually certain that the evening rotation was rostered to close five-sector duties within 45 minutes of the limit. It is likely that this increased risk on the rotation, but the roster was not found to have contributed to this occurrence. (Safety issue)
- Floor escape path lighting was virtually certainly checked only every 24 months, not pre-flight. This gap in monitoring very likely increased risk in any evacuation. It was not found to have contributed to this occurrence or its outcome. (Safety issue)

Other findings

- A tailwind component during the take-off roll and stop was examined and ruled out. Its existence was very unlikely.
- A No. 2 engine power loss before the reject decision was examined and ruled out. It was very unlikely to have existed, and the No. 2 power loss began at impact.
- Maximum braking was virtually certainly established within 2 seconds of the reject decision. This very likely reduced risk by limiting how far the aircraft travelled into the runway end safety area.

Recommendations

Standard's threshold left the aerodrome with no documented wildlife plan

Aerodrome Standard AS-139 section 10.4.2 requires a documented wildlife hazard management plan only where an aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. Tarlton Springs met neither limb. Its heaviest scheduled aeroplane was 22,900 kg, and it handled 61,400 passenger movements in 2025. Under section 10.4.3, its wildlife measures did not need to be documented at all. The aerodrome was therefore not in breach of the standard.

The aerodrome held no plan, no hazard assessment and no dispersal programme. Its wildlife sighting log contained 41 kangaroo entries, but no operator had ever asked to see it.

The issue matters because a documented plan brings a hazard assessment, an owner and an audit against both. Without one, the failed northern fence sat in the works register as an ordinary maintenance item. It was deferred through three budget rounds with no hazard assessment before the council. The absence of a plan likely contributed to the fence remaining unrepaired. That conclusion rests on a narrower balance of evidence than the other contributing factors. Section 10.4.1 still obliged the aerodrome operator to take reasonable measures, and there was no evidence that a documented plan would have secured funding for the repair.

No safety action has been recorded against this issue, and no recommendation is made.

Failed northern fence left unrepaired through three budget rounds

In 04/2023 the aerodrome operator recorded the northern boundary fence between the 700 m and 1,300 m runway markers as failed, with animals observed passing through. Replacement was estimated at \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. By 02/2026 the section had deteriorated further. On 23/06/2026 a continuous 140 m section had failed, with an animal pad through it at the 1,050 m mark (Figure 1).

The failed section was the route by which the two kangaroos reached the movement area. It likely contributed to their reaching the sealed runway. The aerodrome operator had known of the breach for more than three years before the occurrence.

No safety action has been recorded against this issue. The works register called for replacement without stating a specification, and kangaroos can clear a stock fence. [NOT ESTABLISHED: the specification of the replacement fence and whether it will exclude kangaroos.] No recommendation has been recorded against this issue.

Evening service approved without reopening the daylight-only risk assessment

The operator approved a scheduled evening service into Tarlton Springs without reopening the route risk assessment that restricted the aerodrome to daylight operations. RRA-041 issue 3 (11/2022) carried the restriction "daylight operations only pending review of night lighting and diversion fuel policy". CMR-2025-118 introduced the evening rotation from 14/07/2025. It assessed the change as a schedule change only, found no new hazard, and recorded that no route risk assessment review was required. The change record did not mention the restriction. The process asked whether the change was new. It did not ask whether the existing assessment still held.

The safety significance lies in what the restriction would have prevented. Had it been honoured, every company departure from Tarlton Springs would have been made in daylight. Instead, SB512 left on schedule and began its take-off roll at 17:30, after the end of civil twilight at 17:25. It was likely that approving the service without reopening the assessment contributed to the roll being made in darkness.

A review might still have approved the evening service, with conditions on lighting and diversion fuel, because the restriction named those items and not wildlife. The approver could not be interviewed, so what a review would have concluded could not be established. The more likely outcome is that a review assessing a night operation at a single-runway aerodrome would not have approved the service unchanged.

No safety action has been recorded against this issue, and no recommendation is made.

Inspection after last light required only where a wildlife strike was reported in 24 months

The operator required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A of Operations Manual Part A. An aerodrome was listed only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision asked a crew to report a sighting, and no other evidence could admit an aerodrome to the list. Tarlton Springs had never been listed. The aerodrome's own log held eleven sightings of kangaroos on the sealed surface. These were not strikes and were not reported to the company, so they could not bring the requirement into effect.

The inspection requirement itself was sound. The weakness was the listing criterion, which could only be met by the occurrence the control existed to prevent. No aerodrome could therefore be listed before a strike had happened there. The criterion very likely contributed to the occurrence, because no runway inspection was requested before the departure.

No safety action has been recorded against this issue, and no recommendation is made.

No training or guidance for a runway obstruction during the take-off roll

The operator's recurrent training and operations manual gave flight crew no preparation for an obstruction on the runway during the take-off roll. It was extremely likely that this gap existed. All four rejected take-off scenarios in module 4 of TRG-SYL-07 are aircraft system events. No crew in the five training cycles on file had been presented with an external obstruction, and the current cycle alone covers 214 crew. Part B 4.6 directs that the take-off be continued at or above V1. It gives no guidance on an obstruction seen on the runway at any speed, or on a case in which continuing is itself hazardous.

The gap likely contributed to the take-off being rejected 6 kt above V1. The captain made that decision in about a second, without a briefed or practised frame to decide against. He also stated that he knew the rule and departed from it deliberately, because he judged continuing to be worse. On that reading, the gap is a matter for the fleet rather than a cause of this reject. The investigation found the first reading likely. Even so, the gap explains the conditions under which the decision was made. It does not excuse the decision.

The gap reaches beyond this crew. Every crew trained under the current syllabus and manual would face an obstruction during the take-off roll with the same lack of preparation. Both flight crew raised the matter independently.

No safety action has been recorded against this issue.

Evening rotation rostered to close five-sector duties within 45 minutes of the limit

The operator rostered the evening rotation, SB511 Port Carrick to Tarlton Springs and SB512 Tarlton Springs to Port Carrick, as the fourth and fifth sectors of a duty signing on at Port Carrick between 06:00 and 07:00. The rotation operated on 289 days between 14/07/2025 and 18/06/2026. On each of those days the duty was planned at between 12 hours 45 minutes and 13 hours 15 minutes. The maximum for five sectors from those sign-on times was 13 hours 30 minutes.

The roster pattern was unlikely to have contributed to this occurrence. It likely increased risk to the operation all the same. The exposure is built into the roster and is not confined to one day. Every operation of the rotation closes the longest duty in the pattern at an aerodrome that RRA-041 records as having no alternate within 200 nm at night and limited engineering support.

The exposure is sharpest when a rostered crew member reports unfit. The rotation then goes to a crew already on duty, and that crew's day is extended from three sectors to five on the morning of operation. This happened six times in the period, the sixth on 18/06/2026.

No safety action has been recorded against this issue, and no recommendation is made.

Floor escape path lighting checked only every 24 months, not pre-flight

On this type, the pre-flight emergency lighting test illuminated the ceiling emergency lights and the exit signs. It did not illuminate or monitor the floor proximity escape path segments. The only check of those segments was a dedicated functional test at the maintenance programme's 24-month interval. Between functional tests, nothing available to the crew or a line engineer showed the state of the escape path lighting on any aircraft of the type.

On VH-SBQ the gap concealed a defect. The first officer noticed sections of the aisle floor strip unlit during the disembarkation. A functional check on 24/06/2026 then found two of the six floor proximity segments aft of row 16 on the left side dark on battery power with the system armed. The fault was traced to a failed common ballast. The last functional test had been 11 months before the occurrence, inside the interval. The defect could have stayed undetected for up to 13 more months.

The gap did not contribute to the overrun or its outcome. No evacuation was commanded, and the main cabin lights were on throughout. It very likely increased risk nonetheless. With a fire indication, smoke or a night evacuation over the wing, dark segments in the aft cabin could be the difference between an orderly exit and a blocked one. [NOT ESTABLISHED: whether the failed ballast part is fitted across the fleet, and how many aircraft of the type are currently inside the functional test interval.]

No safety action has been recorded against this issue, and no recommendation is made.

Distribution

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